Unsupervised Deep Learning

Week 3: Latent Variable Models

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Lecture Overview

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1. Motivation
2. Latent Variable Models
3. Learning with Latents
4. EM Algorithm

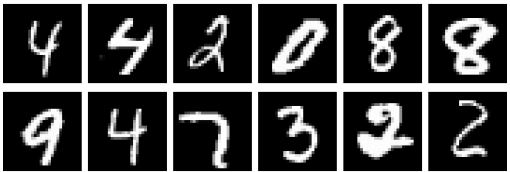
From Autoregressive to Latent

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Autoregressive models:

- Chain of conditionals $p(\mathbf{x}) = p(x_1)\,p(x_2 \mid x_1)\,p(x_3 \mid x_1, x_2) \cdots p(x_D \mid x_1, \ldots, x_{D-1})$
- Sampling one variable at a time
- No explanation of the data - no compact description

Hidden Causes: Latent Variable Model



Data (images) $\mathbf{x}$

Latent factors: (unobserved) $\mathbf{z}$:

- Which digit
- Thickness, slant, size
- Writer, pen

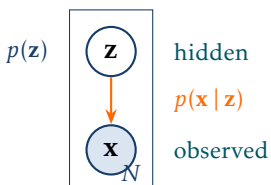
Joint: $p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z})\, p(\mathbf{x} \mid \mathbf{z})$

Marginal:

$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{z}} p(\mathbf{z})\, p(\mathbf{x} \mid \mathbf{z}) =$

Generate:

1. sample $\mathbf{z} \sim p(\mathbf{z})$ - prior
2. sample $\mathbf{x} \sim p(\mathbf{x} \mid \mathbf{z})$ - likelihood



Example: Mixture of Gaussians

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Latent: $z \in \{1, \ldots, K\}$ - group

Prior: $\text{Cat}(\pi_1, \ldots, \pi_K) \Rightarrow p(z = k) = \pi_k$

Likelihood: $p(\mathbf{x} \mid z = k) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k)$

Generate:

1. sample $\mathbf{z} \sim p(\mathbf{z})$ - prior $\Rightarrow$ pick group k
2. sample $\mathbf{x} \sim p(\mathbf{x} \mid \mathbf{z})$ - likelihood $\Rightarrow$, sample $\mathbf{x} \sim \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k)$ from group k

Marginal GMM: $p(\mathbf{x}) = \sum_k p(\mathbf{x}, z = k) = \sum_k p(z = k)\, p(\mathbf{x} \mid \mathbf{z}) = \sum_k \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k)$

Maximum Likelihood with Latents

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Loss: negative log-likelihood of data $\mathcal{D} = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)}\}$, parameters $\boldsymbol{\theta}$

loss: $\mathcal{L}(\boldsymbol{\theta}) = -\sum_{i=1}^{N} \log p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)})$

Simple Gauss model (no z): $p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \Sigma), \quad \boldsymbol{\theta} = \{\boldsymbol{\mu}, \Sigma\}$

loss: $\mathcal{L}(\boldsymbol{\theta}) = -\sum_{i=1}^{N} \log \mathcal{N}(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}, \Sigma)$

solution: log cancels exp in Gaussian pdf $\rightarrow$ closed form for $\boldsymbol{\mu}, \Sigma$

Mixture of Gaussians (hidden z): $p_{\boldsymbol{\theta}}(\mathbf{x}) = \sum_k \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k) = \sum_k p_{\boldsymbol{\theta}}(\mathbf{x}, z = k)$, $\boldsymbol{\theta} = \{\pi_k, \boldsymbol{\mu}_k, \Sigma_k\}_{k=1}^{K}$

loss: $\mathcal{L}(\boldsymbol{\theta}) = -\sum_{i=1}^{N} \log \sum_k \pi_k \mathcal{N}(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k, \Sigma_k)$

solution: log of sum $\rightarrow$ no closed form

$\Rightarrow$ **Gradient descent instead?**

Posterior of the Latent

Gradient of loss:

$$\nabla_\theta \mathcal{L}(\theta) = -\sum_{i=1}^{N} \nabla_\theta \log \sum_k \pi_k \mathcal{N}(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k, \Sigma_k) = -\sum_{i=1}^{N} \nabla_\theta \log \sum_k f_k$$

One example: $f_k = \pi_k \mathcal{N}(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k, \Sigma_k) = p(\mathbf{x}^{(i)}, z = k)$ remember: $\nabla \log u = \nabla u / u$

$$\nabla_\theta \log \sum_k f_k = \frac{\nabla_\theta \sum_k f_k}{\sum_{k'} f_{k'}} = \frac{\sum_k \nabla_\theta f_k}{\sum_{k'} f_{k'}} = \sum_k \frac{f_k}{\sum_{k'} f_{k'}} \nabla_\theta \log f_k$$

Weights: $\dfrac{f_k}{\sum_{k'} f_{k'}} = \dfrac{p(\mathbf{x}^{(i)}, z{=}k)}{\sum_{k'} p(\mathbf{x}^{(i)}, z{=}k')} = \dfrac{p(\mathbf{x}^{(i)}, z{=}k)}{p(\mathbf{x}^{(i)})} = p(z{=}k \mid \mathbf{x}^{(i)})$

Posterior of latent: which z likely produced this $\mathbf{x}$ $\Rightarrow$ **responsibility** r_k

Back to loss: $\nabla_\theta \mathcal{L}(\theta) = -\sum_{i=1}^{N} \sum_k \underbrace{p(z{=}k \mid \mathbf{x}^{(i)})}_{r_k^{(i)}} \nabla_\theta \log \left(\pi_k \mathcal{N}(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k, \Sigma_k) \right)$

Getting the Parts

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Given: current parameters $\boldsymbol{\theta} = \{\pi_k, \boldsymbol{\mu}_k, \Sigma_k\}_{k=1}^{K}$, single example $\mathbf{x}^{(i)}$

1. Joint: $f_k = \pi_k \mathcal{N}(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k, \Sigma_k) = p(\mathbf{x}^{(i)}, z{=}k)$ for $k = 1, \ldots, K$

2. Marginal: $p(\mathbf{x}^{(i)}) = \sum_{k=1}^{K} p(\mathbf{x}^{(i)}, z{=}k) = \sum_{k=1}^{K} f_k$ sum over all values of z

3. Posterior: $r_k^{(i)} = p(z{=}k \mid \mathbf{x}^{(i)}) = f_k \,/\, p(\mathbf{x}^{(i)})$

4. Gradient: $\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \mathbf{x}^{(i)}) = -\sum_k r_k^{(i)} \nabla_{\boldsymbol{\theta}} \log f_k$ has closed form

Discrete z: enumerate K terms per example - tractable and cheap

Idea - iterate: freeze $\boldsymbol{\theta}$ → get $r_k^{(i)}$ → loss closed form → new $\boldsymbol{\theta}$

EM for Mixture of Gaussians

Start: initial parameters $\pi_k, \boldsymbol{\mu}_k, \Sigma_k$. Repeat:

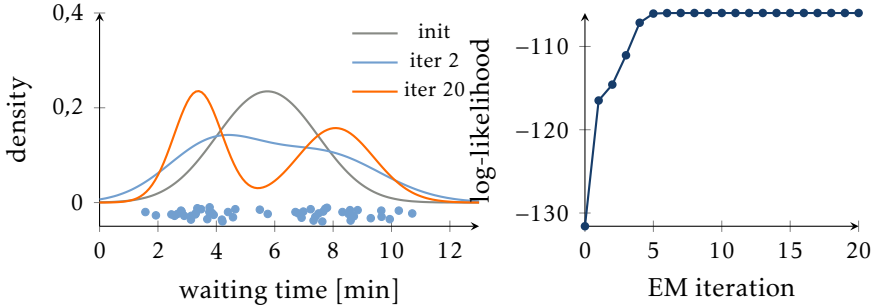
E-step: responsibilities for current parameters

$$r_{ik} = p(z = k \mid \mathbf{x}_i) = \frac{\pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j)}$$

M-step: weighted maximum likelihood, $N_k = \sum_i r_{ik}$

$$\pi_k = \frac{N_k}{N}, \quad \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_i r_{ik} \mathbf{x}_i, \quad \Sigma_k = \frac{1}{N_k} \sum_i r_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top$$

EM in Action



Bad start → two clear bumps in few steps

Log-likelihood: goes up every iteration